Breadth First Search

AI-7.py - C:/Users/Vineela/OneDrive/Documents/AI-7.py (3.15.0)

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```
from collections import deque

graph = {
    "A": ["B", "C"],
    "B": ["D", "E"],
    "C": ["F"],
    "D": [],
    "E": [],
    "F": []
}

def bfs(start):
    visited = set()
    queue = deque([start])

    while queue:
        node = queue.popleft()

        if node not in visited:
            print(node, end=" ")
            visited.add(node)

            for neighbor in graph[node]:
                if neighbor not in visited:
                    queue.append(neighbor)

bfs("A")
```

IDLE Shell 3.15.0

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```
Python 3.15.0 (tags/v3.15.0:d220409, Oct  9 2026, 11:55:56) [MSC v.1951 64 bit (
AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
========== RESTART: C:/Users/Vineela/OneDrive/Documents/AI-7.py ==========
A B C D E F
>>>
```